

Open electronics for a biophysical model for intraspinal mechanosensing in avians

Motivation

- Fast bipedal locomotion requires fast mechanosensing and muscle action (Daley 2006)
- Lengths of sensorimotor pathways add to nerve conduction delay: > 40ms for large animals (More 2010, 2018)
- Lumbosacral morphology and accessory lobes suggest a fast, intraspinal mechanosensor: lumbosacral organ (LSO) (Necker 1999, Eide 1996, Kamska 2020)
- Fossil record also shows the lumbosacral enlargement (Giffin 1990)
- Here we hypothesize that lumbosacral, intraspinal soft tissue oscillates during locomotion (Kamska 2020)
- Problem: lumbosacral soft tissue cannot be observed during locomotion in vivo

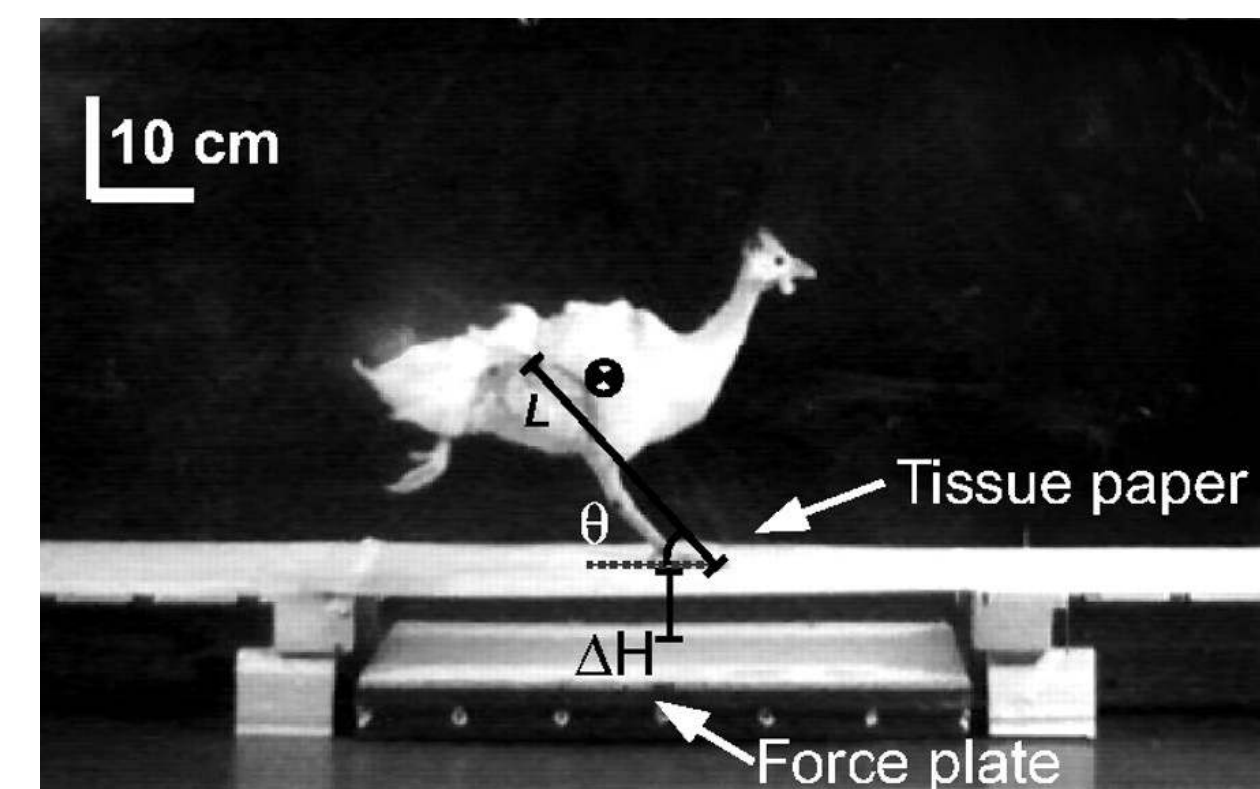
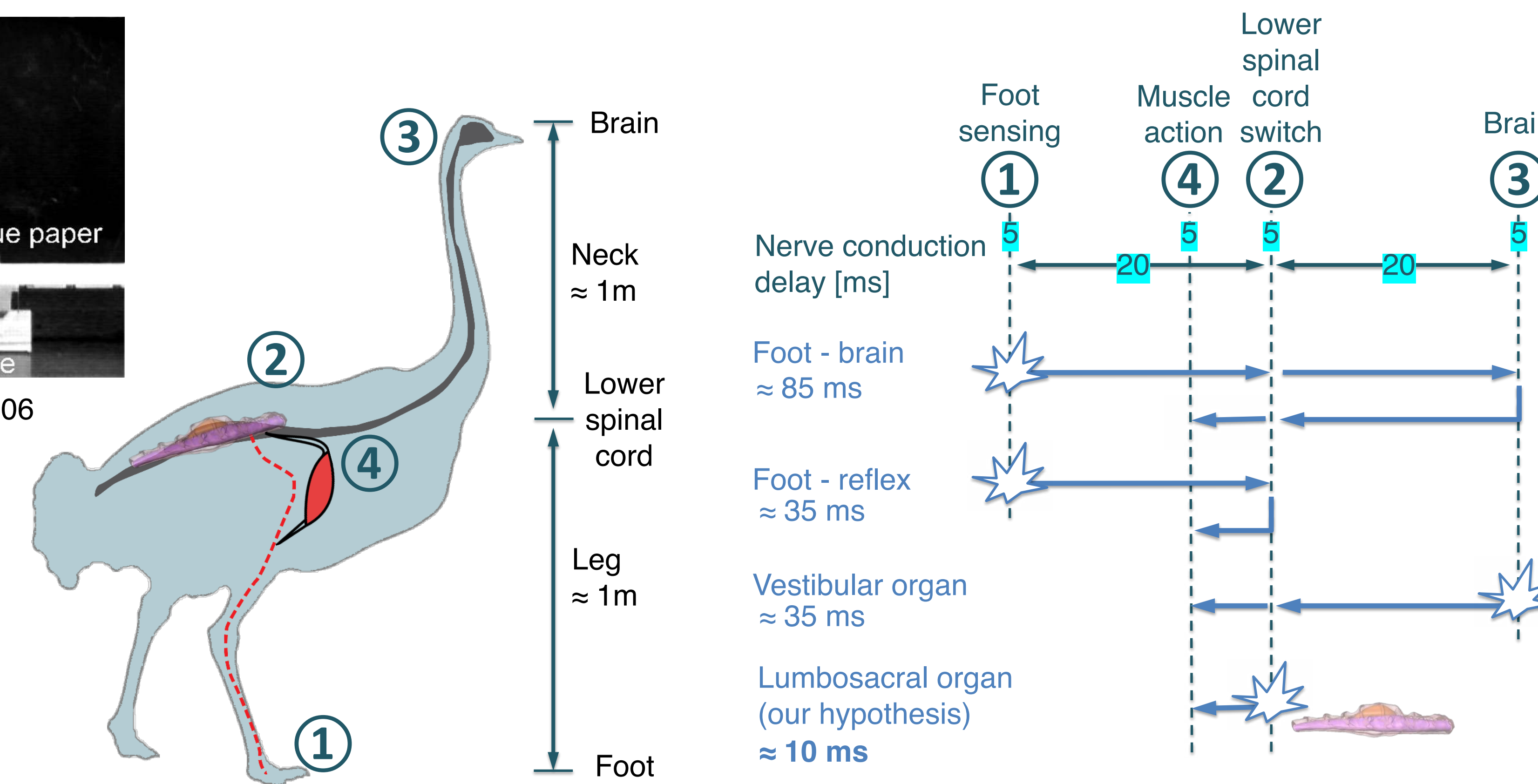


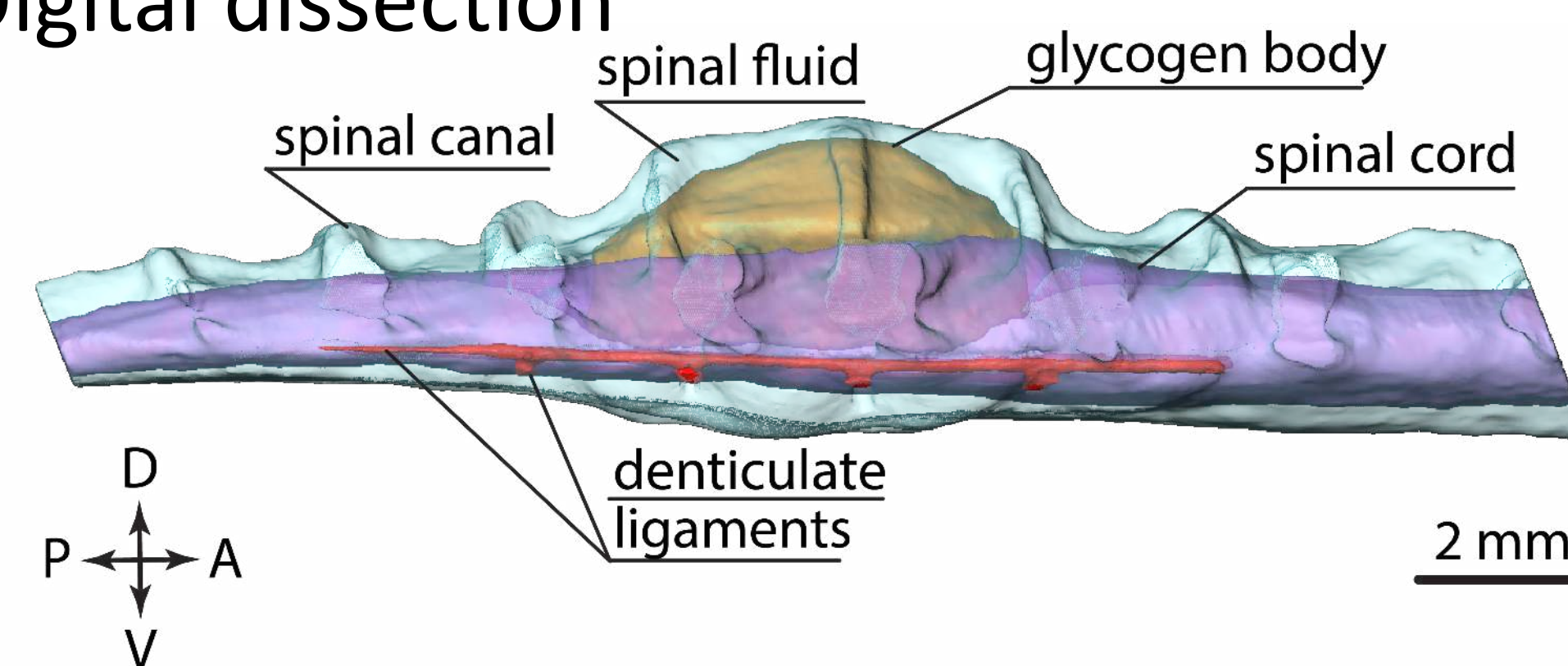
Image taken from Daley 2006



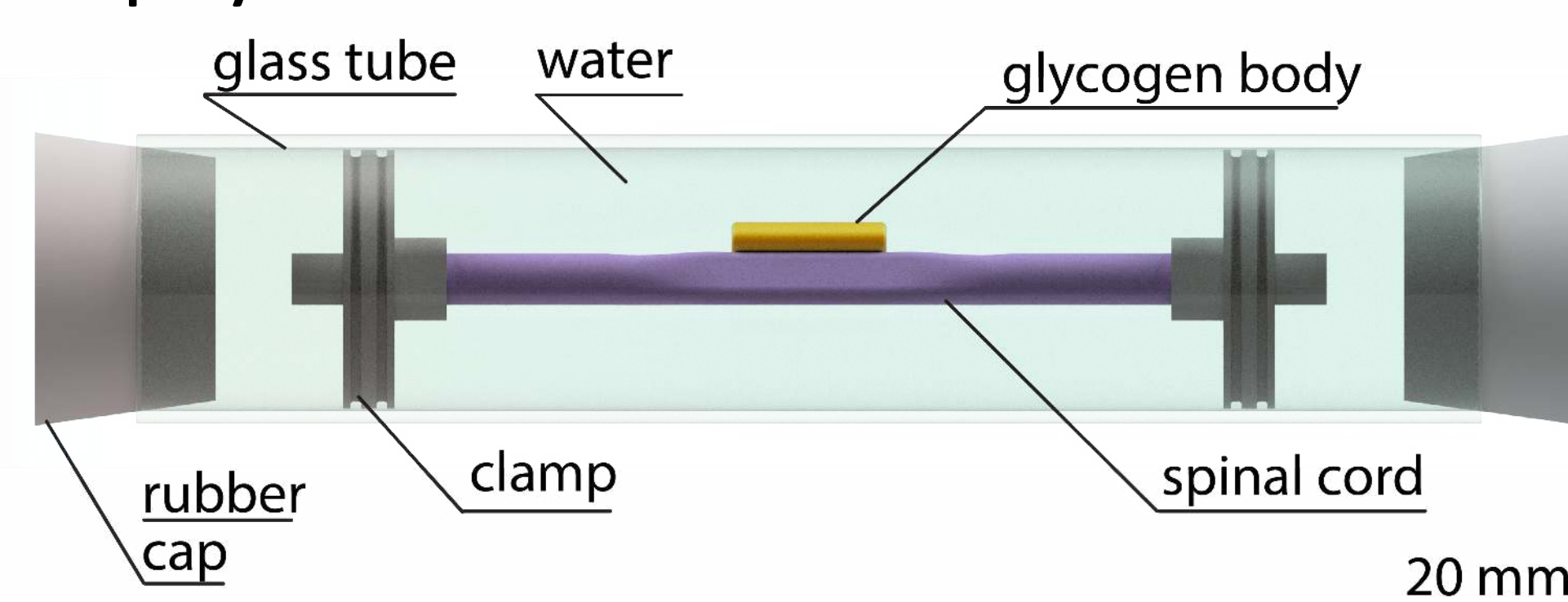
Methods

We built a biophysical model based on our digital dissection observation. We used open electronics to develop a highly dynamic locomotion simulator for mechanically shaking the biophysical model, to simulate forces in locomotion conditions.

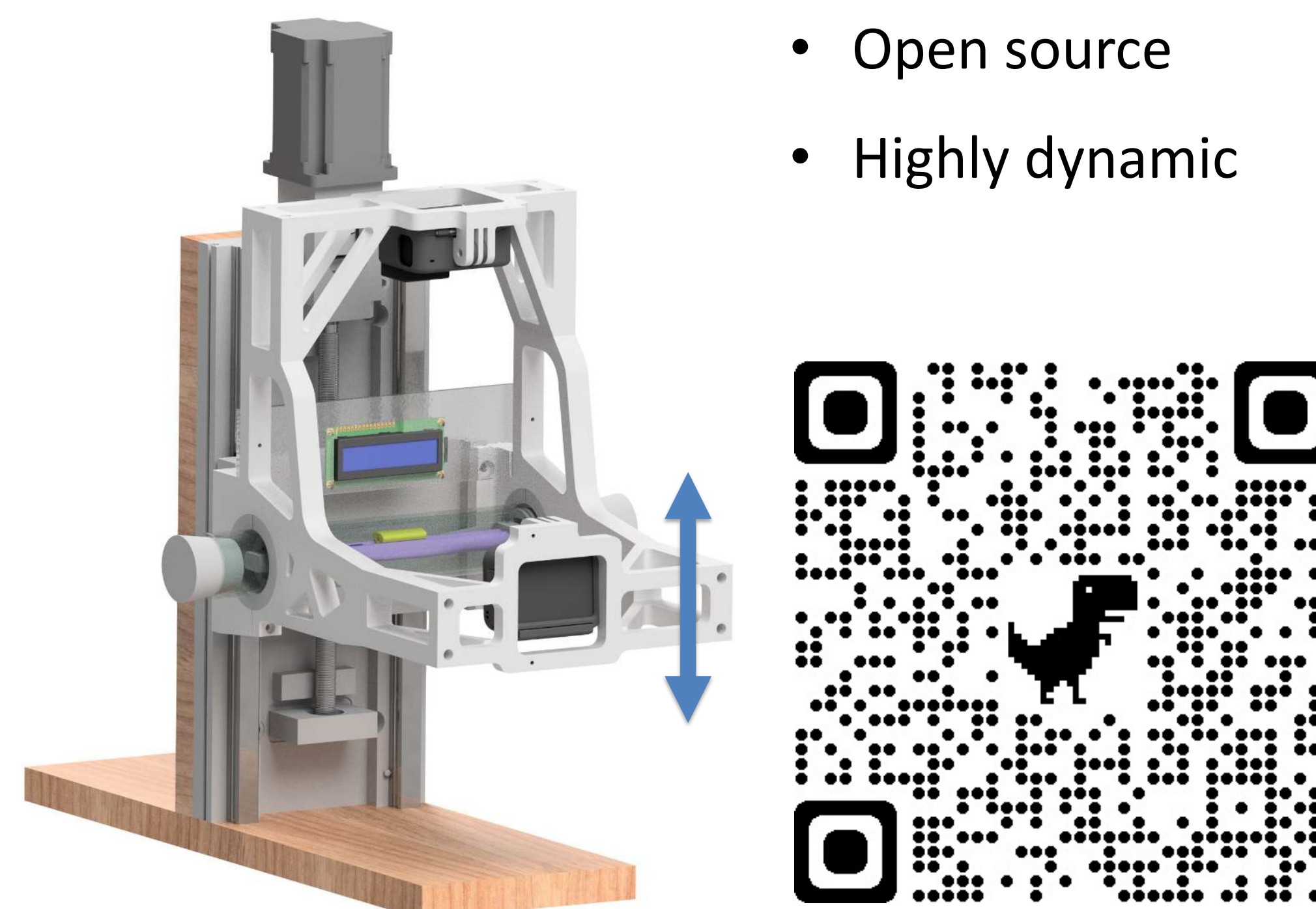
Digital dissection



Biophysical model



Locomotion simulator



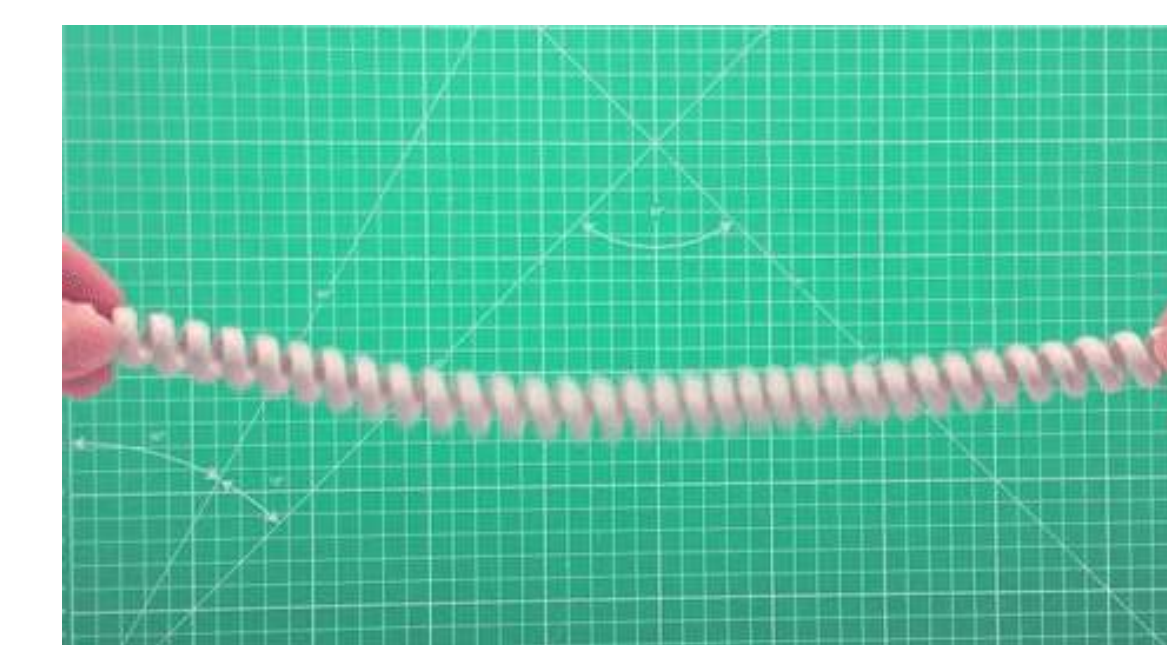
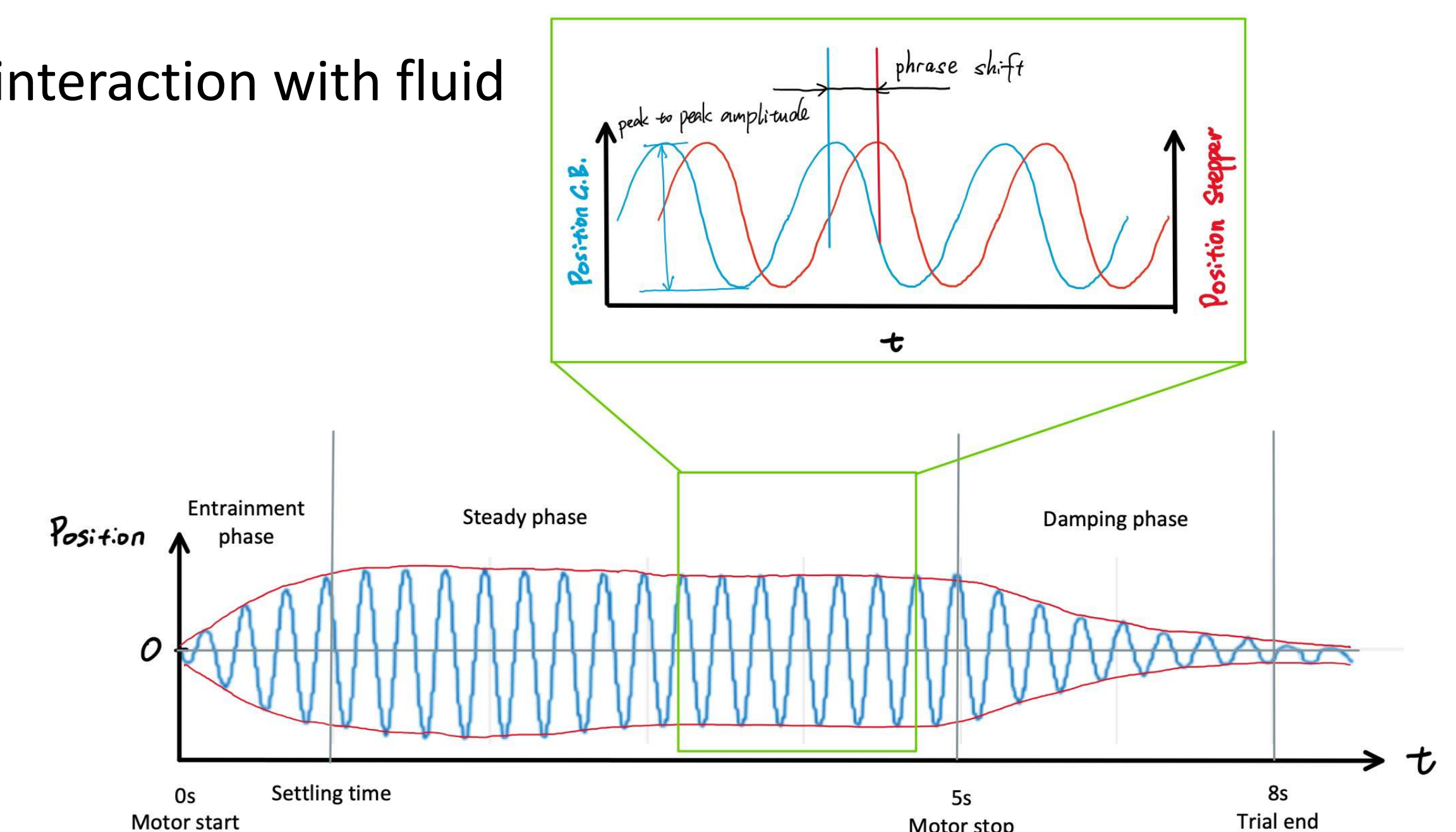
https://github.com/moanan/1_dof_motion_simulator

- Open source
- Highly dynamic

Expected results

We aim to quantify soft tissue motion during locomotion with a biophysical model. We expect to see effects from

- elastic deformation of soft tissue
- soft tissue interaction with fluid



Spinal cord resembles a rubber wire in a snow globe

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